1.34 Differentiation: Stationary Points

Worked Solutions

1a)

$$\frac{dy}{dx} = 2x - 2 = 0 \text{ so } x = 1.$$

1b)

$$\frac{dy}{dx} = -4x + 18 = 0 \text{ so } x = \frac{9}{2}.$$

1c)

$$\frac{dy}{dx} = 3x^2 - 6x + 3 = 0$$
$$3(x^2 - 2x + 1) = 3(x-1)^2 \text{ so } x = 1.$$

1d)

$$y = x - 4x^{-2} \text{ and } \frac{dy}{dx} = 1 + 8x^{-3} = 0$$
$$1 + \frac{8}{x^3} = 0 \Rightarrow x^3 = -8 \text{ so } x = -2.$$

1e)

$$y = x + 4x^{-1} \text{ and } \frac{dy}{dx} = 1 - 4x^{-2} = 0$$
$$1 - \frac{4}{x^2} = 0 \Rightarrow x^2 = 4 \text{ so } x = \pm 2.$$

1f)

$$y = x^2 + 2x^{-1} \text{ and } \frac{dy}{dx} = 2x - 2x^{-2} = 0$$
$$2x - \frac{2}{x^2} = 0 \Rightarrow x^3 = 1 \text{ so } x = 1.$$

1g)

$$y = \frac{1}{3}x + x^{-3}$$
$$\frac{dy}{dx} = \frac{1}{3} - 3x^{-4} = 0$$
$$\frac{1}{3} - \frac{3}{x^4} = 0 \Rightarrow x^4 = 9 \text{ so } x = \pm\sqrt{3}.$$

2a)

$$f'(x) = 2x + 4 < 0 \text{ so } x < -2$$
$$\{x : x \in \mathbb{R}, x < -2\}.$$

2b)

$$f'(x) = 12x - 10 < 0 \text{ so } x < \frac{5}{6}$$
$$\left\{x : x \in \mathbb{R}, x < \frac{5}{6}\right\}.$$

2c)

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1) < 0$$

so $-1 < x < 1$.

$$\{x : x \in \mathbb{R}, -1 < x < 1\}.$$

2d)

$$\begin{aligned} f'(x) &= 6x^2 + 6x - 36 = 6(x^2 + x - 6) \\ &= 6(x+3)(x-2) < 0 \end{aligned}$$

so $-3 < x < 2$.

$$\{x : x \in \mathbb{R}, -3 < x < 2\}.$$

2e)

$$\begin{aligned} f'(x) &= -6x^2 - 36x - 48 = -6(x^2 + 6x + 8) \\ &= -6(x+4)(x+2) < 0 \end{aligned}$$

so $x < -4$ or $x > -2$.

$$\{x : x \in \mathbb{R}, x < -4\} \cup \{x : x \in \mathbb{R}, x > -2\}.$$

3)

$$\begin{aligned} f'(x) &= -3x^2 - 12x + 15 = -3(x^2 + 4x - 5) \\ &= -3(x+5)(x-1) > 0 \end{aligned}$$

so $-5 < x < 1$.

$$x \in (-5, 1).$$

4)

$$\begin{aligned} f'(x) &= 3x^2 - 6x + 4 = 3(x^2 - 2x) + 4 \\ &= 3[(x-1)^2 - 1] + 4 = 3(x-1)^2 + 1. \end{aligned}$$

As $(x-1)^2 \geq 0$, $3(x-1)^2 + 1 > 0$ for all x, so $f(x)$ always increasing.

5)

$$\begin{aligned} f'(x) &= -9x^2 + 6x - 8 = -(9x^2 - 6x + 8) \\ &= -\left(9\left(x^2 - \frac{2}{3}x\right) + 8\right) \\ &= -\left(9\left(\left(x - \frac{1}{3}\right)^2 - \frac{1}{9}\right) + 8\right) \\ &= -\left(9\left(x - \frac{1}{3}\right)^2 - 1 + 8\right) \\ &= -9\left(x - \frac{1}{3}\right)^2 - 7. \end{aligned}$$

As $\left(x - \frac{1}{3}\right)^2 \geq 0$, $-9\left(x - \frac{1}{3}\right)^2 - 7 < 0$ for all x, so $f(x)$ always decreasing.

6a)

$$\frac{dy}{dx} = -2x + 5 \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$-2x + 5 = 0 \text{ so } x = \frac{5}{2}, \quad y = -\left(\frac{5}{2}\right)^2 + 5 \times \frac{5}{2} - 6 = \frac{1}{4}$$

$$\frac{d^2y}{dx^2} = -2 \text{ so } \left.\frac{d^2y}{dx^2}\right|_{x=\frac{5}{2}} = -2 < 0$$

$\therefore \left(\frac{5}{2}, \frac{1}{4}\right)$ is a maximum.

6b)

$$\frac{dy}{dx} = 6x^2 - 30x + 24 \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$6(x-1)(x-4) = 0 \text{ so } x = 1 \text{ or } x = 4$$

$$y|_{x=1} = 2 - 15 + 24 = 11, \quad y|_{x=4} = 128 - 240 + 96 = -16$$

$$\frac{d^2y}{dx^2} = 12x - 30$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=1} = -18 < 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=4} = 18 > 0$$

$\therefore (1, 11)$ is a maximum and $(4, -16)$ is a minimum.

6c)

$$\frac{dy}{dx} = 6x^2 - 6x - 12 \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$6(x+1)(x-2) = 0 \text{ so } x = -1 \text{ or } x = 2$$

$$y|_{x=-1} = -2 - 3 + 12 + 7 = 14, \quad y|_{x=2} = 16 - 12 - 24 + 7 = -13$$

$$\frac{d^2y}{dx^2} = 12x - 6$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = -18 < 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=4} = 18 > 0$$

$\therefore (-1, 14)$ is a maximum and $(2, -13)$ is a minimum.

6d)

$$\frac{dy}{dx} = -3x^2 + 6x + 72 \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$-3(x+4)(x-6) = 0 \text{ so } x = -4 \text{ or } x = 6$$

$$y|_{x=-4} = 64 + 48 - 288 = -176, \quad y|_{x=6} = -216 + 108 + 432 = 324$$

$$\frac{d^2y}{dx^2} = -6x + 6$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-4} = 30 > 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=6} = -30 < 0$$

$\therefore (-4, -176)$ is a minimum and $(6, 324)$ is a maximum.

6e)

$$\frac{dy}{dx} = 8x^3 - 8x \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$8x(x^2 - 1) = 8x(x+1)(x-1) = 0$$

so $x = -1$ or $x = 0$ or $x = 1$.

$$y|_{x=-1} = 2 - 4 = -2, \quad y|_{x=0} = 0, \quad y|_{x=1} = 2 - 4 = -2$$

$$\frac{d^2y}{dx^2} = 24x^2 - 8$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = 16 > 0, \quad \left.\frac{d^2y}{dx^2}\right|_{x=1} = -8 < 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=1} = 16 > 0$$

$\therefore (-1, -2)$ is a minimum, $(0, 0)$ is a maximum and $(1, -2)$ is a minimum.

6f)

$$\frac{dy}{dx} = 4 - x^{-2} \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$4 - \frac{1}{x^2} = 0 \Rightarrow 4 = \frac{1}{x^2} \Rightarrow \frac{1}{4} = x^2 \Rightarrow x = \pm\frac{1}{2}$$

$$y|_{x=-\frac{1}{2}} = -2 - 2 = -4, \quad y|_{x=\frac{1}{2}} = 2 + 2 = 4$$

$$\frac{d^2y}{dx^2} = -2x^{-3}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-\frac{1}{2}} = -16 < 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=\frac{1}{2}} = 16 > 0$$

$\therefore \left(-\frac{1}{2}, -4\right)$ is a maximum, $\left(\frac{1}{2}, 4\right)$ is a minimum.

6g)

$$\frac{dy}{dx} = -2x^{-3} + 2 \quad \text{stat pt when } \frac{dy}{dx} = 0$$
$$-\frac{2}{x^3} + 2 = 0 \Rightarrow 2 = \frac{2}{x^3} \Rightarrow 1 = x^3 \Rightarrow x = 1$$
$$y|_{x=1} = 1 + 2 = 3$$
$$\frac{d^2y}{dx^2} = 6x^{-4} \text{ so } \left.\frac{d^2y}{dx^2}\right|_{x=1} = 6 > 0$$

$\therefore (1, 3)$ is a minimum.

6h)

$$\frac{dy}{dx} = 18x^5 - 18x^2 \quad \text{stat pt when } \frac{dy}{dx} = 0$$
$$18x^2(x^3 - 1) = 8x(x^2 + x + 1)(x - 1) = 0$$

so $x = 0$ or $x = 1$.

$$y|_{x=0} = 0, \quad y|_{x=1} = 3 - 6 = -3$$
$$\frac{d^2y}{dx^2} = 90x^4 - 36x$$
$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=1} = 54 > 0$$
$$\left.\frac{dy}{dx}\right|_{x=-0.1} = 18x^5 - 18x^2 = -0.180818$$
$$\left.\frac{dy}{dx}\right|_{x=0.1} = 18x^5 - 18x^2 = -0.17982$$

$\therefore (0, 0)$ is a stationary point of inflection and $(1, -3)$ is a minimum.

7)

$$\frac{dy}{dx} = -9x^2 - 4x$$
$$-9x^2 - 4x = -9\left(x^2 + \frac{4}{9}x\right)$$
$$= -9\left(\left(x + \frac{2}{9}\right)^2 - \frac{4}{81}\right)$$
$$= -9\left(x + \frac{2}{9}\right)^2 + \frac{4}{9}.$$

When $x = -\frac{2}{9}$, the maximum of the gradient function is $\frac{4}{9}$. This occurs at $\left(-\frac{2}{9}, \frac{227}{243}\right)$.

8a)

$$\frac{dy}{dx} = 3x^2 - 6x - 45 \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$3(x^2 - 2x - 15) = 3(x-5)(x+3) = 0$$

so $x = -3$ or $x = 5$.

$$y|_{x=-3} = -27 - 27 + 135 = 81, \quad y|_{x=5} = 125 - 75 - 225 = -175$$

$$\frac{d^2y}{dx^2} = 6x - 6$$

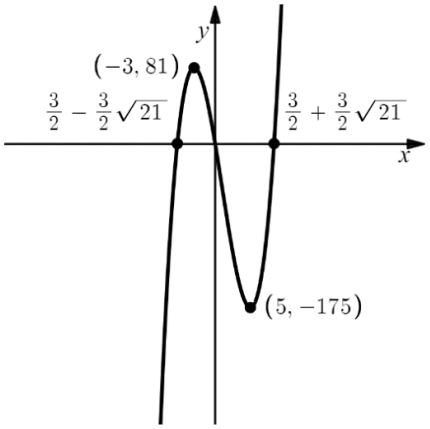
$$\left.\frac{d^2y}{dx^2}\right|_{x=-3} = -24 < 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=5} = 24 > 0$$

$\therefore (-3, 81)$ is a maximum and $(5, -175)$ is a minimum.

When $x = 0$, $y = 0$. When $y = 0$,

$$x^3 - 3x^2 - 45x = x(x^2 - 3x - 45) = 0$$

so $x = \dfrac{3 \pm 3\sqrt{21}}{2}$.



8b)

$$\frac{dy}{dx} = 64x^3 - 64x \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$64x(x^2 - 1) = 0 \text{ so } x = -1,\ x = 0 \text{ or } x = 1$$

$$y|_{x=-1} = 16 - 32 + 7 = -9, \quad y|_{x=0} = 7, \quad y|_{x=1} = 16 - 32 + 7 = -9$$

$$\frac{d^2y}{dx^2} = 192x^2 - 64$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=-1} = 128 > 0, \quad \left.\frac{d^2y}{dx^2}\right|_{x=0} = -64 < 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=1} = 128 > 0$$

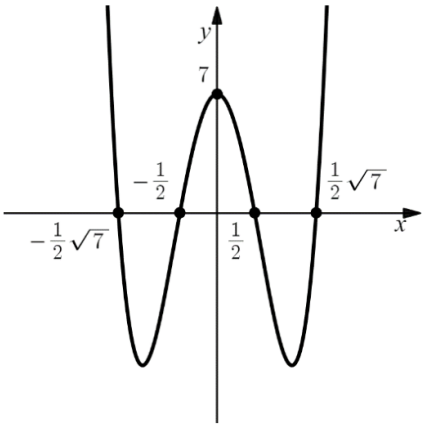
$\therefore (-1, -9)$ is a minimum, $(0, 7)$ is a maximum and $(1, -9)$ is a minimum.

When $x = 0$, $y = -1$.

When $y = 0$, $16x^4 - 32x^2 - 7 = 0$.

$$(2x - 1)(2x + 1)(4x^2 - 7) = 0$$

so $x = \frac{1}{2}$, $x = -\frac{1}{2}$ and $x = \dfrac{\pm\sqrt{7}}{2}$.



8c)

$$\frac{dy}{dx} = 5 - x^{-\frac{1}{2}} \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$5 - \frac{1}{\sqrt{x}} = 0 \Rightarrow \sqrt{x} = \frac{1}{5} \text{ so } x = \frac{1}{25}$$

$$y|_{x=\frac{1}{25}} = \frac{1}{5} - \frac{2}{5} = -\frac{1}{5}$$

$$\frac{d^2y}{dx^2} = \frac{1}{2}x^{-\frac{3}{2}}, \quad \left.\frac{d^2y}{dx^2}\right|_{x=\frac{1}{25}} = \frac{1}{150} > 0$$

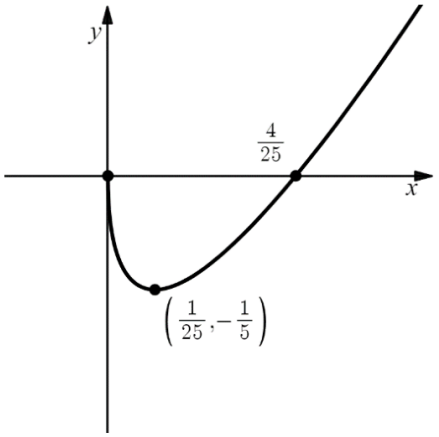
$\therefore \left(\frac{1}{25}, -\frac{1}{5}\right)$ is a minimum.

When $x = 0$, $y = 0$.

When $y = 0$, $5x - 2\sqrt{x} = 0$.

$$\Rightarrow x\left(5 - \frac{2}{\sqrt{x}}\right) = 0 \text{ so } x = 0$$

or $5 = \frac{2}{\sqrt{x}} \Rightarrow \sqrt{x} = \frac{2}{5} \Rightarrow x = \frac{4}{25}$.



9)

$$\frac{dy}{dx} = -3x^2 + 2ax + b$$

Stationary point: $\frac{dy}{dx} = 0$ at $x = 1$.

$$0 = -3 + 2a + b$$

Stationary point: $\frac{dy}{dx} = 0$ at $x = 3$.

$$0 = -27 + 6a + b$$

So $a = 6$, $b = -9$.

At $(2, 4)$,

$$4 = -8 + 24 - 18 + c \text{ so } c = 6.$$

10)

$$y = ax^4 + bx^2 + c \text{ so } \frac{dy}{dx} = 4ax^3 + 2bx$$

$$\frac{d^2y}{dx^2} = 12ax^2 + 2b$$

$(1, 0)$ gives $0 = a + b + c$.

$(0, -3)$ gives $c = -3 \Rightarrow a + b = 3$.

$(\sqrt{2}, 1)$ gives $8\sqrt{2}a + 2\sqrt{2}b = 0$.

$$\Rightarrow 4a + b = 0 \text{ so } a = -1,\ b = 4$$

$$y = -x^4 + 4x^2 - 3$$

$$\frac{dy}{dx} = -4x^3 + 8x = 0 \Rightarrow -4x(x^2 - 2) = 0$$

So stat pts at $x = 0, -\sqrt{2}$ and $\sqrt{2}$.

$$\frac{d^2y}{dx^2} = -12x^2 + 8$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = 8, \quad \left.\frac{d^2y}{dx^2}\right|_{x=-\sqrt{2}} = -16 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{x=\sqrt{2}} = -16$$

$x = 0, -\sqrt{2}$ and $\sqrt{2}$.

So maximum at $(-\sqrt{2}, 1)$, minimum at $(0, -3)$ and maximum at $(\sqrt{2}, 1)$.

11)

$$f'(x) = 2e^{2x} - 1 \quad \text{increasing when } f'(x) > 0$$

$$2e^{2x} - 1 > 0 \Rightarrow e^{2x} > \frac{1}{2} \Rightarrow 2x > \ln\frac{1}{2} \Rightarrow x > \frac{1}{2}\ln\frac{1}{2}$$

or $x > -\frac{1}{2}\ln 2$.

So $x \in \left(-\frac{1}{2}\ln 2, \infty\right)$.

12a)

When $x = 0$, $y = 5 - 1 = 3 \ \therefore (0, 4)$.

When $y = 0$, $5e^x - e^{2x} = 0$.

$$e^x(5 - e^x) = 0 \text{ so } e^x = 5 \Rightarrow x = \ln 5 \ \therefore (\ln 5, 0).$$

12b)

$$\frac{dy}{dx} = 5e^x - 2e^{2x} \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$5e^x - 2e^{2x} = 0 \Rightarrow e^x(5 - 2e^x) = 0$$

$e^x \neq 0$ so $5 - 2e^x = 0$.

$$\Rightarrow e^x = \frac{5}{2} \Rightarrow x = \ln\left(\frac{5}{2}\right)$$

$$y = 5e^{\ln(\frac{5}{2})} - e^{2\ln(\frac{5}{2})} = \frac{25}{2} - \frac{25}{4} = \frac{25}{4}$$

So stat point at $\left(\ln\left(\frac{5}{2}\right), \frac{25}{4}\right)$.

12c)

$$\frac{d^2y}{dx^2} = 5e^x - 4e^{2x}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=\ln(\frac{5}{2})} = -\frac{25}{2} < 0$$

so maximum at $\left(\ln\left(\frac{5}{2}\right), \frac{25}{4}\right)$.

13)

$$\frac{dy}{dx} = 3e^{3x} - 2e^{2x} - e^x \quad \text{stat pt when } \frac{dy}{dx} = 0$$

$$3e^{3x} - 2e^{2x} - e^x = 0$$

$$\Rightarrow e^x(3e^{2x} - 2e^x - 1) = 0$$

$$\Rightarrow e^x(3e^x + 1)(e^x - 1) = 0$$

$e^x \neq 0$ and $3e^x + 1 \neq 0$ so $e^x = 1 \Rightarrow x = 0$.

$$y = 1 - 1 - 1 = -1$$

$$\frac{d^2y}{dx^2} = 9e^{3x} - 4e^{2x} - e^x$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=0} = 4 > 0$$

so minimum at $(0, -1)$.